Julie Keisler

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Education

PhD in Computer Science *Feb 2022 – Feb 2025*
University of Lille - EDF R&D - INRIA Lille & Paris

- Automated Deep Learning: algorithms and software for energy sustainability.
- Supervised by Pr. Claire Monteleoni, Pr. El-Ghazali Talbi and Margaux Brégère.

Engineer Degree (MS in Computer Science) *Sept 2018 – Sept 2021*
Télécom Paris

- Major in signal processing for AI, minor in computer graphics and interactive systems.

Exchange student *Sept 2020 – Feb 2021*
ETH Zürich

- Semester Thesis in the Power System Laboratory: Benchmark electric power consumption forecasting algorithms, supervised by Yi Wang.
- **Coursework:** Power Market - Portfolio and Risk Management, Neural Network Theory, Environmental Systems Data Science, Corporate Sustainability.

CPGE, MPSI/MP* *Sept 2016 – Sept 2018*
Lycée Montaigne, Bordeaux

- Undergraduate studies to prepare for competitive entry exams to french engineering schools (Grandes Ecoles). Subjects studied: Mathematics, Computer Science, Physics, French literature, German and English.

Awards

INREC 2024 conference, best paper award *Sept 2024*
Automated Spatio-Temporal Weather Modeling for Load Forecasting., with Margaux Brégère

Flood Map Prediction challenge, winning team. *Jan - April 2024*

- Prediction of flood risk maps without streamflow data.
- Winning proposition with a CNN-based solution, team work with Eva Girousse.
- Workshop paper at ICLR 2025 Workshop: Tackling Climate Change with Machine Learning.

Research Experience

Starting research position *June 2025 - present*
INRIA Paris - EDF R&D

- Generative artificial intelligence for statistical climate downscaling.
- Joint work with Pr. Claire Monteleoni, Anastase Charantonis, Yannig Goude and Boutheina Oueslati.

Data Scientist Intern - 6 months *Feb - August 2021*
Sagemcom R&D

- Data Analysis and load forecasting for a rural electrification project in Madagascar.
- Development of microservices for customers (Docker, Jenkins, Spark technologies).

Teaching Experience

Teaching Assistant *2024 - Present*
Faculté des Sciences d'Orsay - Université Paris Saclay

- Practical Work: Introduction to Deep Learning, for the MS Mathematics and AI.
- Developed seven practical assignments on deep learning and optimization, covering convolutional networks, architecture and hyperparameter search, GANs, diffusion models, NLP, and GNNs.

Research Supervisor *2024 - Present*

Supervised Master students interns

- **Manon Tavernier** (AgroParisTech & ENS Paris-Saclay): Temporal dissagregation for statistical climate downscaling, INRIA Paris.
- **Haidar Ali Yousef** (Mines Paris- PSL & Lebanese University): Statistical climate downscaling with generative models, INRIA Paris.
- **Elyas Chikhaoui** (Télécom Paris): Symbolic regression for satellite data using DRAGON, INRIA Paris.
- **Keshav Das** (Télécom Paris): Automated selection of adaptive additive models, application to load consumption forecasting, EDF R&D.
- **Alban Derepas** (ENSAE): Future evolution of the wind resource and the interest of machine learning methods for statistical wind downscaling, EDF R&D & INRIA Paris.
- **Roxane Goffinet** (Télécom Paris): Global forecasting models for a large number of time series, EDF R&D.

Selected Publications

Journal Articles

WindDragon: Automated deep learning for regional wind power forecasting. *March 2025*

Julie Keisler, Etienne Le Naour

Environmental Data Science

An algorithmic framework for the optimization of deep neural networks architectures and hyperparameters. *June 2024*

Julie Keisler, El-Ghazali Talbi, Sandra Claudel, Gilles Cabriel

Journal of Machine Learning Research (JMLR)

Peer-reviewed Conference Proceedings

AutoML algorithms for online generalized additive model selection: application to electricity demand forecasting *Sept 2025*

Keshav Das, ***Julie Keisler***, Amaury Durand, Margaux Brégère

International Conference on Automated Machine Learning (AutoML)

Automated Deep Learning for Load Forecasting. *Sept 2024*

Julie Keisler, Sandra Claudel, Gilles Cabriel, Margaux Brégère

International Conference on Automated Machine Learning (AutoML)

Peer-reviewed without Proceedings

A hackathon for flood map prediction from geospatial data with parsimonious machine learning models. *April 2025*

David Medernach, Cyril Lemaire, Eva Girousse, ***Julie Keisler***, Julie Richon, Nicolas Brunel

ICLR 2025 Workshop on Tackling Climate Change with Machine Learning

Automated Spatio-Temporal Weather Modeling for Load Forecasting *Sept 2024*

Julie Keisler, Margaux Brégère

INREC 2024 conference, Best Paper Award

A Bandit Approach with Evolutionary Operators for Model Selection. *Aug 2024*

Margaux Brégère, ***Julie Keisler***

International Workshop on Resource-Efficient Learning for Knowledge Discovery, ACM SigKDD

WindDragon: Enhancing wind power forecasting with Automated Deep Learning. *April 2024*

Julie Keisler Etienne Le Naour,
ICLR 2024 Workshop on Tackling Climate Change with Machine Learning

Submitted

SerpentFlow: Generative Unpaired Domain Alignment via Shared-Structure Decomposition. *Jan 2026*

Julie Keisler, Anastase Charantonis, Boutheina Oueslati, Yannig Goude, Claire Monteleoni

Preprint

Generative Unsupervised Downscaling of Climate Models via Domain Alignment: Application to Wind Fields. *Jan 2026*

Julie Keisler, Boutheina Oueslati, Anastase Charantonis, Yannig Goude, Claire Monteleoni

Preprint

Various Research Activities

Software

Python package DRAGON *Documentation*

- Python package for the optimization of deep neural network architectures and hyperparameters.

Service & Dissemination

External Service

- Conference Reviewer: ICANN24, CCAI workshop at NeurIPS24.
- Journal Reviewer: Environmental Data Science.
- CIFRE testimony presentation during the Forum Maths et Entreprise (FEM) at CNAM.
- Student mentorship and tutoring with Télécom Paris and TechPourToutes.
- Invited lecture on AI for weather and climate for civil servants at the Institut des Hautes Études, Ministère de l'Intérieur.

INRIA Service

- Lead weekly team meetings.
- Development and maintenance of the team website.

Selected Talks

PEPR Climat - TRACCS, Assemblée Générale 2026, PC10 Localising *Paris, February 2026*

- SerpentFlow, unpaired downscaling of wind fields

CERFACs and Meteo-France Seminar *Toulouse, December 2025*

- Downscaling CMIP6 wind speed using Deep Learning.

Kayrros and INRIA seminar *Paris, October 2024*

- Downscaling CMIP6 wind speed to ERA5 resolution using Deep Learning.

PhD Defense *Palaiseau, January 2024*

- Automated Deep Learning: Algorithms and Software for Energy Sustainability.

ENBIS-24 *Leuven, September 2024*

- A Bandit Approach With Evolutionary Operators for Model Selection: Application to Neural Architecture Optimization for Image Classification.

AutoML 2024 *Paris, September 2024*

- Automated Deep Learning for load forecasting.

INREC 2024 *Essen, September 2024*

- Automated Spatio-Temporal Weather Modeling for Load Forecasting.

SIGKDD Workshop on Resource-Efficient Learning for Knowledge Discovery *Barcelona, August 2024*

- A Bandit Approach With Evolutionary Operators for Model Selection: Application to Neural Architecture Optimization for Image Classification.

55es Journées de Statistique de la SFdS *Bordeaux, May 2024*

- Mutant-UCB: entre bandits et algorithme évolutionnaire, une approche pour la sélection de modèles.

Workshop series on applied Machine Learning in Energy and Climate Science. *Online, May 2024*

- Automated Deep Learning for load forecasting.

ICLR 2024 Workshop on Tackling Climate Change with Machine Learning *Wien, May 2024*

- WindDragon: enhancing wind power forecasting with automated deep learning.

SCOOL Seminar (INRIA Lille). *Lille, November 2023*

- Optimizing Deep Neural Networks Architectures and Hyperparemeters using bandits algorithms.

ConfStochStatML workshop, Wolfgang Pauli Institute. *Wien, Sept 2023*

- Short-term load forecasting using optimized Deep Neural Networks.

54es Journées de Statistiques de la SFdS. *Brussels, July 2023*

- Algorithme de bandits pour l'optimisation des hyperparametres de réseaux de neurones.

PGMO Days. *Palaiseau, November 2022*

- Evolving directed acyclic graphs for Automated Deep Learning: application to short-term load forecasting.

OLA'2023 Conference. *Syracuse, June 2022*

- Deep Transformers Optimization for load forecasting.

Skills

Programming Languages: Python, R, C++

Packages: PyTorch, TensorFlow, Scikit-Learn, Numpy, Pandas, Xarray.

Tools: Linux, Latex, Git, Slurm on HPC, MPI, Docker.

Languages: French (native), English (fluent), German (intermediate, AbiBac: Abitur with 1.0 in 2016).

Others: Football, guitar (end of musical studies certificate), volunteer for Data For Good.